Patient Profile

Name: Saubhik Bhaumik
Age: Not Provided
Gender: Male

Summary of Results

All evaluated liver function parameters fall within established reference ranges, indicating normal hepatic synthetic and excretory function at this time.

Detailed Test Explanations

- **Serum Bilirubin (Total)** – 0.9 mg/dL (Normal: 0.2–1.2). Measures overall bilirubin load; normal levels indicate adequate hepatic conjugation and excretion.
- **Serum Bilirubin (Direct)** – 0.2 mg/dL (Normal: 0–0.3). Reflects conjugated bilirubin; normal values suggest efficient liver conjugation and biliary clearance.
- **Serum Bilirubin (Indirect)** – 0.7 mg/dL (Normal: 0.2–1.0). Represents unconjugated bilirubin; within range confirms adequate hemolysis control and hepatic uptake.
- **SGPT (ALT)** – 36 U/L (Normal: 13–40). Enzyme released with hepatocellular injury; normal value indicates no significant hepatocyte damage.
- **SGOT (AST)** – 32 U/L (Normal: 0–37). Similar to ALT; normal level supports intact liver cell integrity.
- **Serum Alkaline Phosphatase** – Not Available. This enzyme assesses cholestasis and bone activity; absence precludes interpretation for this marker.
- **Serum Protein** – 7.2 g/dL (Normal: 6.4–8.3). Total plasma proteins; normal range signifies standard albumin and globulin synthesis.
- **Serum Albumin** – 4.7 g/dL (Normal: 3.5–5.2). Key marker of hepatic synthetic function and nutritional status; within normal limits.
- **Globulin** – 2.5 g/dL (Normal: 1.8–3.6). Represents immune globulins; normal value aligns with healthy immune and hepatic production.
- **AIG Ratio** – 1.88 (Normal: 1.1–2.4). Ratio of Albumin to Globulin; normal proportion indicates balanced protein synthesis.

Overall Interpretation

The comprehensive assessment demonstrates unremarkable liver function. Enzymatic indices of hepatocellular integrity (ALT, AST) and bilirubin metabolism are within normal limits, suggesting absence of active hepatic injury, cholestasis, or hemolytic overload. The protein panel confirms adequate synthetic capacity.

Recommendations

1. **Maintain a balanced diet** rich in fruits, vegetables, lean proteins, and whole grains to support liver health.
2. **Limit alcohol consumption** and avoid self-medication with hepatotoxic substances.
3. **Engage in regular physical activity** (moderate aerobic exercise) to promote metabolic homeostasis.
4. **Monitor liver function periodically** (annual testing or as advised) especially if medication use or alcohol intake changes.
5. **Maintain adequate hydration** and avoid prolonged fasting to reduce hepatic metabolic stress.

Disclaimer

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This report is for informational purposes only and does not replace professional medical evaluation or individualized clinical advice. Consultation with a qualified healthcare provider is recommended for personalized interpretation and management.